Piyush Patle

+91 9284910474 | piyushpatle228@gmail.com | linkedin.com/in/piyush-patle
github.com/PiyushPatle26 | piyushpatle26.github.io

Education

Veermata Jijabai Technological Institute — Mumbai, India
B.Tech in Electrical Engineering, Minor in Robotics — *Aug 2023 - May 2027*

Open-Source Contributions

Linux Kernel | *Linux Foundation Kernel Mentorship Program (LKMP)* Merged Patches | lore — Spring 2026
- Picked up TODO items from the kernel's task lists and upstreamed them: device tree bindings converted to YAML schema, Tegra210 byte-map code simplified, two drivers moved to `FIELD_GET()`, and a Kbuild documentation fix.
- Added the AVIA HX710B to the `hx711` driver by reworking it to handle more than one chip and extending its binding; groundwork merged and queued for v7.3, device patches on v12 in review.

Apache NuttX | *TI AM62x architecture port* Merged PRs
- Wrote the first AM62x port in NuttX under `arch/arm64`: boot code, memory map, IRQ definitions, early console, and a 16550 lower half, plus BeaglePlay and PocketBeagle 2 board support.
- Brought the UART up by fixing FIFO setup, TX polling, and leftover bootloader console state, and the EL2-to-EL1 handoff so EL1 could use the GICv3 system registers.
- Added a TISCI client over the Secure Proxy mailbox for power, reset, and clock control, then GPIO and I2C drivers on top, validated on BeaglePlay and PocketBeagle 2 hardware.

Embox RTOS | *Drivers, board support, toolchain fixes* Merged PRs
- Wrote a SPI NOR flash driver with a JEDEC chip table (SST25, W25Q, MX25L, Spansion, Micron) covering fast read, page program, and erase, tested against an SST25VF016B under QEMU after fixing the i.MX6 ECSPI driver, which forced SPI mode 3 where NOR needs mode 0.
- Closed tracker issues covering unaligned access in LAN9118 and a NOMMU build failure in USB EHCI, and added aarch64 templates that boot Embox on PocketBeagle 2 and BeaglePlay.

Experience

Systems Intern, Mecha | *Vadodara* — May 2026 - Aug 2026
- Brought Linux up on the NXP i.MX95, rebasing upstream patch series to enable HDMI, camera, VPU, and audio.
- Debugged the firmware, device tree, and kernel config failures behind them, validating pipelines with `modetest`, `media-ctl`, `libcamera`, and GStreamer.

Undergrad Research Intern, C-DAC | *Pune* LOR — Jun 2025 - Aug 2025
- Evaluated NFS, BeeGFS, and Lustre for RISC-V HPC, then brought Lustre up on a 4-node StarFive VisionFive 2 (RV64GC) cluster by resolving `ldiskfs/e2fsprogs` and kernel-header build failures.
- Benchmarked I/O and metadata performance with IOR and MDTest, reaching 267 MiB/s read and 123 MiB/s write at 4 clients, competitive with ARM.
- Co-authored the resulting ACM paper on porting and evaluating Lustre for RISC-V HPC (SCA / HPC Asia '26).

Projects

PageForge | *C, RISC-V, no libc* Repository — Apr - Aug 2026
- Wrote a userspace model of the Linux memory management stack in C with no libc: a buddy page allocator over a 4 MB arena, a slab cache for 8 B to 1 KB objects, and `kmalloc/kfree/calloc/realloc` on raw `mmap` and `munmap`.
- Simulated the RISC-V Sv39 page walk in software, covering three-level lookups, address translation, and the permission checks that decide which fault an access would take.
- Cross-compiled for rv64 and validated with 49 Unity tests under `qemu-riscv64` on every commit.

EMG-Armband | *Embedded C, ESP32, ESP-NOW* Report | Blog — Jul - Oct 2024
- Built an EMG acquisition system on ESP32 that streams muscle signals over ESP-NOW in real time, with gesture recognition for device control.

Technical Skills

Languages	C, C++, RISC-V Assembly
Architectures	RISC-V, ARM64
System & Tools	Linux Kernel, NuttX, Embox, Device drivers, QEMU, GDB, `dt_binding_check`, Valgrind
Hardware	TI AM62x, NXP i.MX95/i.MX6, BeagleBoard, StarFive VisionFive 2, STM32, ESP32, Raspberry Pi

Positions of Responsibility

Core Member, Society of Robotics and Automation (SRA), VJTI (Aug 2024 - Present) · ran workshops on Embedded C, ESP32 robotics, OpenCV and ROS 2, and mentored juniors on low-level projects.